



DATA MODELLING

REPRESENTING THE WORLD

1

Data Modelling

2

Workload

3

Relational model

WHAT IS DATA MODELING?

- It is the task to describe (to model) part of the real world in a way that allow you to
 - Storage data (write data)
 - Query data (read data)
- A data model is composed of a conceptual model (a theory) and a language (also a graphic one) for describing and querying data
- A data model is always available in a database management systems that supports the data model semantic
- Example
 - Relational model → MySQL, Oracle, IBM db2, mariaDB..
- What are the interesting parts to model?
- What is the "best" model for my goal?

WORKLOAD

- There are many type of software applications that require different interactions with database
- Write once read many
 - Example?
- Write many ready once
 - Example?
- Write many read many
 - Example?
- Who writes and who reads (employee/tool, customer)

HOW MANY DATA MODELS EXIST?

- Thursday 4 there will be a lesson U4-4 from 13.30 to 16.30 the topic could be the introduction to the course, the type of data and so on
- Fri 5 there is another lesson in the room U4-2 from 15.30 to 17.30 where it is possible to begin to introduce the CAP theorem

tabular

graphical

Orario data management												
Data	giorno	Aula e ora	LEZ	ES	LAB	Learning object						
04-ott	Gio	U4-4 13.30-16.30		3		intro progetti data type						
05-ott	Ven	U4-42 15.30-17.30		2		nosqlcap						
11-ott	Gio	U4-4 13.30-16.30		3		key value with example coloumn						
12-ott	Ven	U4-2 15.30-17.30		2		column example mongogb						
18-ott	Gio	U4-4 13.30-16.30		3		mongodb with example graph						
19-ott	Ven	U4-2 15.30-17.30		2		data replication fragment						
23-ott	mar	14.30-1730				3 virtual lab						
25-ott	Gio	U4-4 13.30-16.30		3		use the right model for the right problem						
26-ott	Ven	U4-2 15.30-17.30		2		mongodb query						
30-ott	mar	14.30-1730			3	mongo						
08-nov	Gio	U4-4 13.30-16.30		3		data distribution relational and not						
09-nov	Ven	U4-2 15.30-17.30		2		neo						
15-nov	Gio	U4-4 13.30-16.30		3		hadoop+data lake						
16-nov	Ven	U4-2 15.30-17.30		2		arango						
22-nov	Gio				3	neo+arango						
23-nov	Ven	U4-2 15.30-17.30		2		hbase						
29-nov	Gio	U4-4 13.30-16.30		3		spark+data integration						
30-nov	Ven	U4-2 15.30-17.30		2		nifi						
06-dic	Gio	U4-4 13.30-16.30		3		kafka						
13-dic	Gio				3	hbase+hadoop						
14-dic	Ven	U4-2 15.30-17.30		2		data integration + quality						
20-dic	Gio				3	kafka+nifi(data quality)						
21-dic	Ven	U4-2 15.30-17.30		2		spark(data quality)						
10-gen	Gio				3	spark(data quality)						
11-gen	Ven	U4-2 15.30-17.30		2		data lifecycle						

	lunedì 08/10	martedì 09/10	mercoledì 10/10	giovedì 11/10	venerdì 12/10
08:00					
08:30	Data technology Batini Carlo, Maurino Andrea				Large scale data management Maurino Andrea, Venturini Claudio
09:00	U24-DISCO-AulaC2 [U24 - Viale Sarca 15, Milano]				U24-DISCO-AulaC2 [U24 - Viale Sarca 15, Milano]
09:30					
10:00	08:30 - 10:30 Lezione				08:30 - 10:30 Lezione
10:30	Data technology Batini Carlo, Maurino Andrea		Large scale data management Maurino Andrea, Venturini Claudio		
11:00	U24-DISCO-AulaC2 [U24 - Viale Sarca 15, Milano]				
11:30	10:30 - 11:30 Lezione		U14-DISCO-Locale T023 [U14 - Viale Sarca 336, Milano]		
12:00			10:30 - 12:30 Lezione		
12:30			Large scale data management Maurino Andrea, Venturini Claudio		
13:00			U14-DISCO-Locale T023 [U14 - Viale Sarca 336, Milano]		

WHAT ARE THE MOST IMPORTANT FEATURES OF A DATA MODEL?

- Machine readability
- Expressive power
- Simplicity
- Flexibility
- Standardization



THE RELATIONAL MODEL



THE RELATIONAL MODEL

- The relational model is a logical model
 - Independent of the specific implementation
- It is based on data modeling in table form and on data connected by values
- What is the query language of the relational model?
 - Among others: relational algebra, SQL (Structured query language)

THE RELATIONAL MODEL

Student

Badge	Surname	Name	Date of Birth
6554	Rossi	Mario	05/12/1998
8765	Neri	Paolo	03/11/1996
9283	Verdi	Luisa	12/11/1999
3456	Rossi	Maria	01/02/1998

Same information
described differently

Exam

Student	Grade	Course
3456	30	04
3456	24	02
9283	28	01
6554	26	01

Same information
Duplicated values (is it good?)
Minimization principle

Badge	Surname	Name	Date of Birth	Course	Grade
3456	Rossi	Maria	01/02/1998	04	30
3456	Rossi	Maria	01/02/1998	02	24
6554	Rossi	Mario	05/12/1998	01	26
9283	Verdi	Luisa	03/11/1996	01	28
8765	Neri	Paolo	03/11/1996		

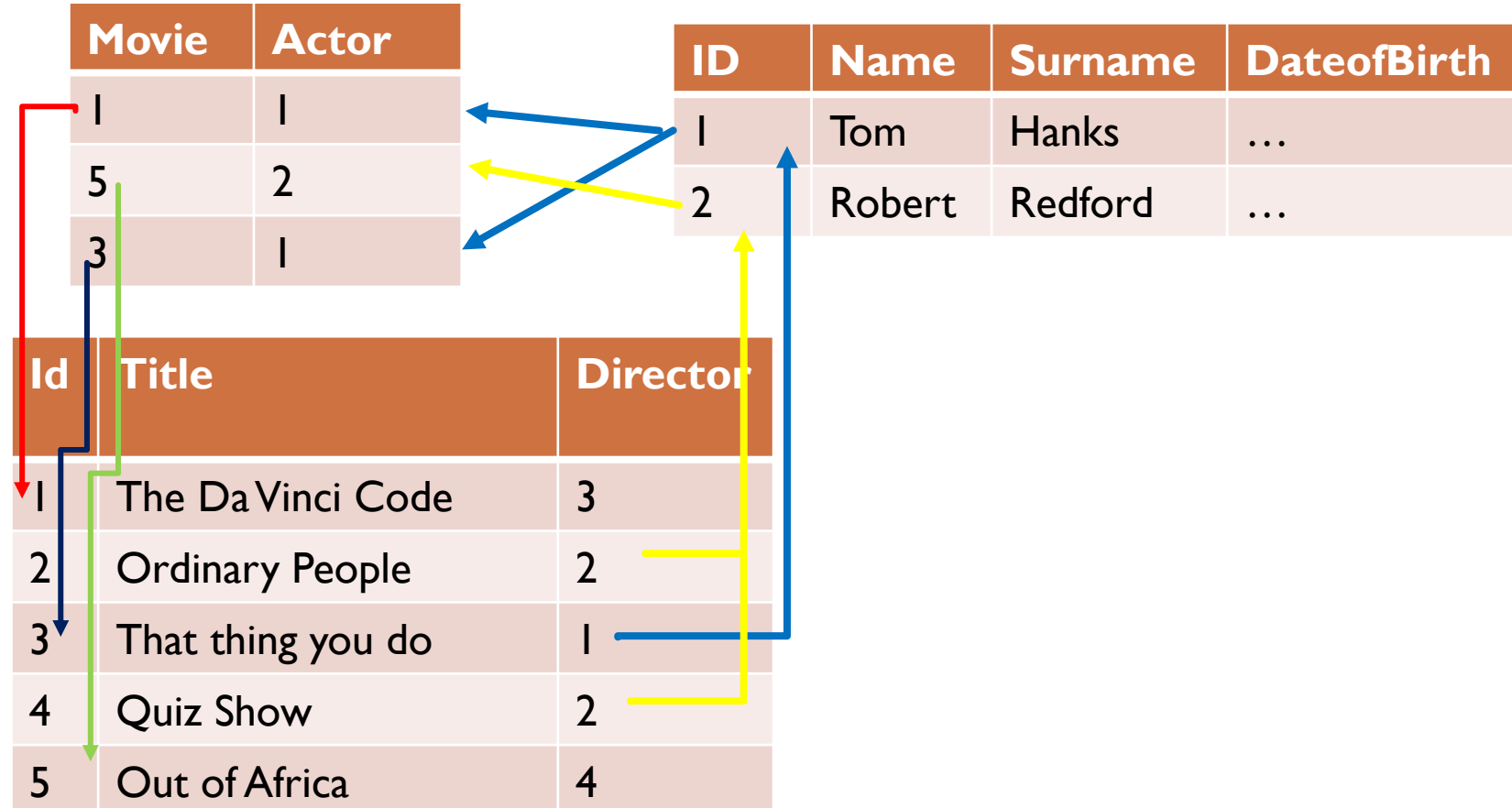
No information

THE RELATIONAL MODEL

- In the relational model «no information» is an information
 - With a different meanings
 - I do not know the value, but I know that it exists
 - The value is not applicable
 - Both
- In all cases where no information is available, the «null» value is used
 - It is not a string it is a logical value as true or false
- Relational model is based on the closed world assumption
- In the relational model the data is linked by values
- In general, each table describes one «entity»
- How many ways are there to connect an entity?
- Movie → director (usually one)
- Director → Movie (usually many)
 - One to many relationships
- Movie → actor (in general many and also vice versa)
 - Many to many

RELATIONAL MODEL

- In case of many to many relationship there is the need to add a new table between entities



HOW TO USE THE RELATIONAL MODEL

- The relational model is implemented in Relational database management systems
 - mySQL, Oracle...
- The table model can also be described with other tools
 - Excel, google sheet (the checkered digital notebook), in some cases proprietary
 - CSV (comma separated values) can be read by any text editor (eg Word, notepad, but also excel)
- How to query them?

Risultati Data management & visualization													
File Modifica Visualizza Inserisci Formato Dati Strumenti Componenti aggiuntivi Guida L'ultima modifica è avvenuta 8 minuti fa													
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fx													
	A	B	C	D	E	F	G	H	I	J	K	L	M
1			43,02%	34,88%	4,65%	3,49%	5,81%	87,21%	81,40%	80,23%	80,23%	79,07%	79,07%
2	media		24,80	25,28	25,00	19,33	23,60	25,14	28,33	27,32	27,32	27,62	27,84
3	totali		86	37	30	4	3	5	75	70	69	68	68
4	matricc	Cognome	1 app DM	2app DM	3app DM	4app DM	5 app DM	voto scritto l	progetto DM	Voto DM	data viz	somma pesa	voto finale
7	834332	ANDREOSE Eugenio	22,5					22,5	25	24,25	24	24,125	24
11	718327	BOWEN Sebastiano		25				25	33	30,6	25	27,8	28
16	779221	BUGGIO Tiziano		25				25	27	26,4	29	27,7	28
17	793129	CALDARA Gabriele	28					28	30	29,4	26	27,7	28

Header

App,Category,Rating,Reviews,Size,Installs,Type,Price,Content Rating,Genres,Last Updated,Current Ver,Android Ver

Photo Editor & Candy Camera & Grid & Scrapbook,ART_AND_DESIGN,4.1,159,19M,"10,000+",Free,0,Everyone,Art & Design,"January 7, 2018",1.0.0,4.0.3 and up

Coloring book moana,ART_AND_DESIGN,3.9,967,14M,"500,000+",Free,0,Everyone,Art & Design;Pretend Play,"January 15, 2018",2.0.0,4.0.3 and up

"U Launcher Lite - FREE Live Cool Themes, Hide Apps",ART_AND_DESIGN,4.7,87510,8.7M,"5,000,000+",Free,0,Everyone,Art & Design,"August 1, 2018",1.2.4,4.0.3 and up

Sketch - Draw & Paint,ART_AND_DESIGN,4.5,215644,25M,"50,000,000+",Free,0,Teen,Art & Design,"June 8, 2018",Varies with device,4.2 and up

Pixel Draw - Number Art Coloring Book,ART_AND_DESIGN,4.3,967,2.8M,"100,000+",Free,0,Everyone,Art & Design;Creativity,"June 20, 2018",1.1,4.4 and up

Paper flowers instructions,ART_AND_DESIGN,4.4,167,5.6M,"50,000+",Free,0,Everyone,Art & Design,"March 26, 2017",1.0,2.3 and up

Smoke Effect Photo Maker - Smoke Editor,ART_AND_DESIGN,3.8,178,19M,"50,000+",Free,0,Everyone,Art & Design,"April 26, 2018",1.1,4.0.3 and up

Infinite Painter,ART_AND_DESIGN,4.1,36815,29M,"1,000,000+",Free,0,Everyone,Art & Design,"June 14, 2018",6.0,61.1,4.2 and up

Garden Coloring Book,ART_AND_DESIGN,4.4,13791,33M,"1,000,000+",Free,0,Everyone,Art & Design,"September 20, 2017",2.9.2,3.0 and up

Kids Paint Free - Drawing Fun,ART_AND_DESIGN,4.7,121,3.1M,"10,000+",Free,0,Everyone,Art & Design;Creativity,"July 3, 2018",2.8,4.0.3 and up

Text on Photo - Fontee,ART_AND_DESIGN,4.4,13880,28M,"1,000,000+",Free,0,Everyone,Art & Design,"October 27, 2017",1.0.4,4.1 and up

Name Art Photo Editor - Focus n Filters,ART_AND_DESIGN,4.4,8788,12M,"1,000,000+",Free,0,Everyone,Art & Design,"July 31, 2018",1.0.15,4.0 and up

Tattoo Name On My Photo Editor,ART_AND_DESIGN,4.2,44829,20M,"10,000,000+",Free,0,Teen,Art & Design,"April 2, 2018",3.8,4.1 and up